

OnLocation – Gallagher Bridge install and config guide

Overview

The OnLocation Gallagher Bridge application interfaces OnLocation person and induction records with Gallagher Command Centre cardholder records. Access to OnLocation is read-only, while access to Gallagher Command Centre is read-write, therefore it is important to keep in mind that while OnLocation is relatively safe from data corruption using this interface, Command Centre is not. If improperly configured it is possible to end up with a loss of information against existing cardholders in Command Centre, or the addition of cardholders which are not required – filling the Command Centre database with unnecessary records. A Command Centre configuration backup prior to completing the initial sync is very important, ensuring a straight forward rollback path in case the bridge application is not correctly setup. A still safer approach would be to replicate the production Command Centre system and first integrate with the isolated copy to verify the integration works as expected, before then implementing with the production instance of Command Centre.

Requirements

The bridge application is fairly lightweight and runs as a Windows Service using the Network Service account. Administrator access is required to install, upgrade and manage the service. The bridge application can be installed on separate machine, VM or the Gallagher server. The bridge service communicates with the OnLocation API endpoint over https connection, so an external TCP path over port 443 is required. Connection to the Gallagher server is via port 8904, configurable from both the Command Centre Web Services configuration, and the bridge application Network Settings section. The default web UI connection is via port 5000, https default (http configurable). Firewall rules are created during installation, but if non-default ports are configured the Windows firewall may need manual update to allow traffic over the new ports.

The bridge application will run on Windows 10/11 or Server 2019 onward, 64 bit OS versions. For detailed information about the application structure and requirements see the GitHub readme here: <https://github.com/ScionResearch/OnLocation-Gallagher-Bridge>

Installation

Download the installer from the ScionResearch GitHub page here: <https://github.com/ScionResearch/OnLocation-Gallagher-Bridge/releases/latest>

Run the installer on the target machine, you will need administrator privileges to complete the installation successfully.

On successful installation run the status widget – this can be found in the Windows Start menu, look for “Status Widget”. The widget appears in the system tray, verify the system status – the indicator on the icon should be amber and the status should be running, not configured (note, if you are upgrading a previously running version, status should be running, fully configured).

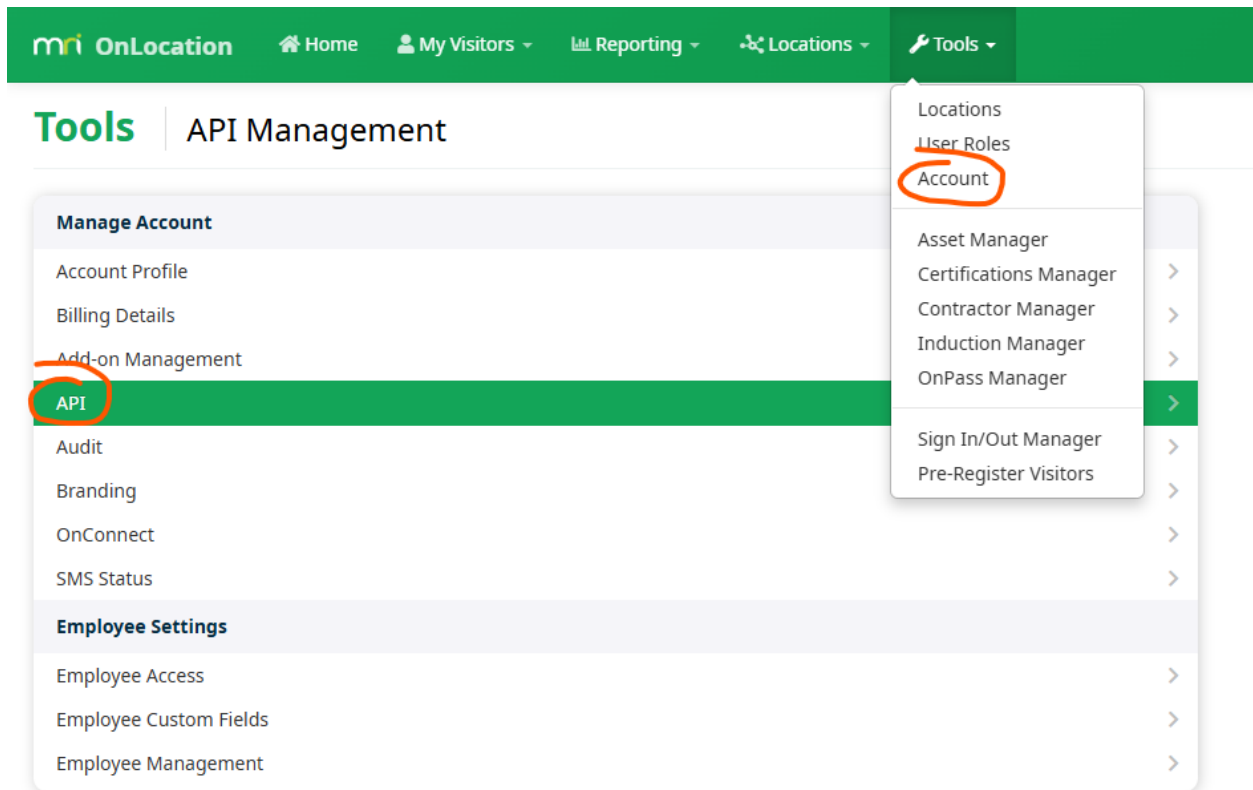
OnLocation configuration

Login to OnLocation here: <https://login.whosonlocation.com/login>

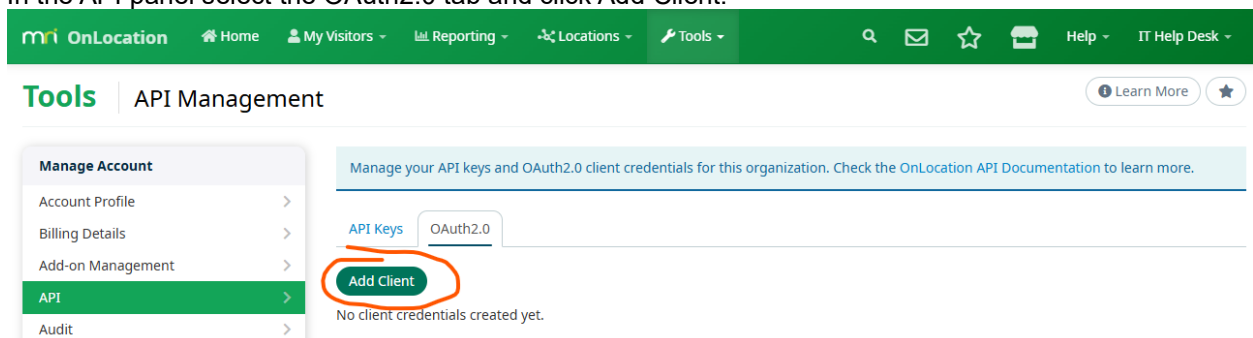
You will need the site administrator credentials in order to create a new API credential – this may be held by the IT helpdesk, or the designated OnLocation support person.

Creating and OAuth2.0 API Key

The OAuth2.0 API Key is the preferred connection method, being the newer – more secure connection method. To configure access via OAuth2.0 - in the OnLocation web portal go to the Tools menu, select Account, and click API in the left hand navigation menu:



In the API Keys tab click the Add Key button, select Unrestricted from the drop-down menu and agree to the Application Developer and API Agreement, then click Generate Key:
In the API panel select the OAuth2.0 tab and click Add Client:



Enter a client name at the top of the dialog box, then select **Inductions**, **Contractor** and **Staff** in the Read column only, and click Create Client

Add Client Credential

Scopes

Domain	Read	Write
Audit Logs	☐	☐
Certifications	☐	☐
Community	☐	☐
Inductions	☒	☐
Locations/Access Point/Departments	☐	☐
Notifications	☐	☐
Contractor	☒	☐
Staff	☒	☐
Visitors	☐	☐

Select one or more scopes. Secrets are only shown once when created or rotated.

[Create Client](#) [Cancel](#)

The Client ID and Client Secret are shown, the Secret will be required for the next step in setting up the bridge application, copy it to a secure document for use in the bridge application setup later, and ensure you delete it once the connection is complete. The secret cannot be accessed once the Client info box is dismissed, but a new one can be generated against the client if necessary. Once you have generated the key, skip to the Gallagher REST API Configuration section.

Creating an API key – Legacy

At the time of writing, basic API keys are supported but marked as legacy. **It is recommended to use the OAuth2.0 method rather than the lower security, legacy key method.** To create a new API key for the bridge application to connect with, in the OnLocation web portal go to the Tools menu, select Account, and click API in the left hand navigation menu:

OnLocation Home My Visitors Reporting Locations Tools

Tools | API Management

Manage Account

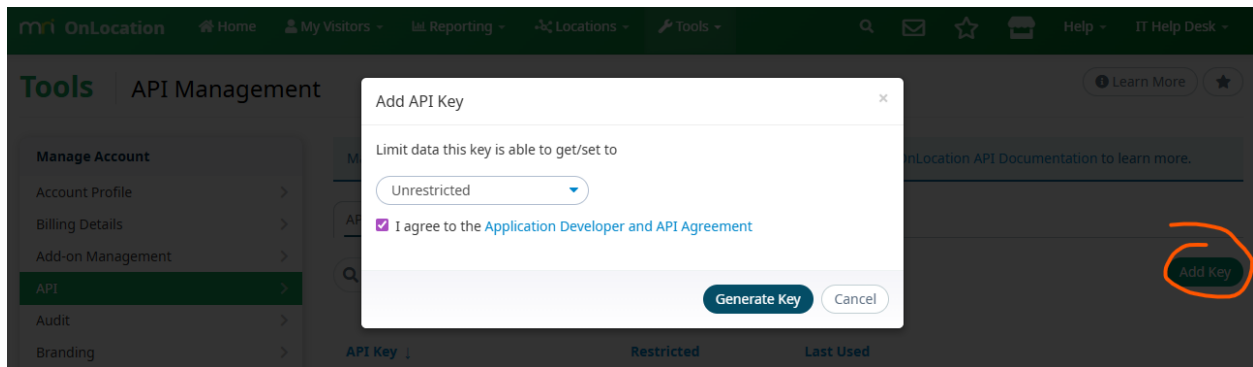
- Account Profile
- Billing Details
- Add-on Management
- API**
- Audit
- Branding
- OnConnect
- SMS Status

Employee Settings

- Employee Access
- Employee Custom Fields
- Employee Management

Locations
User Roles
Account
Asset Manager
Certifications Manager
Contractor Manager
Induction Manager
OnPass Manager
Sign In/Out Manager
Pre-Register Visitors

In the API Keys tab click the Add Key button, select Unrestricted from the drop-down menu and agree to the Application Developer and API Agreement, then click Generate Key:



Note: Unrestricted must be selected for the legacy API key, as the other options do not allow sufficient access to all the records required by the integration.

Gallagher REST API Configuration

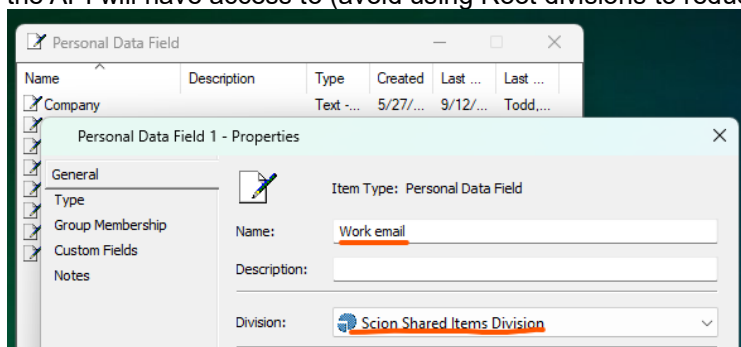
Gallagher Command Centre has several steps required to ensure the interface can access and make changes to the relevant cardholder records. The RESTCardholders licence feature must be enabled – this can be verified from the Server Properties → Licensing → Features tab. Look for RESTCardholders – Enabled to confirm the feature is licensed. Steps to configure are:

1. Create Personal Data Fields
2. Create Competencies for inductions
3. Create a Division for new cardholders
4. Create Access Groups for default access, and personal data field visibility
5. Create a REST API operator configuration and user to define permissions
6. Enable REST API and apply certificate
7. Create REST Client

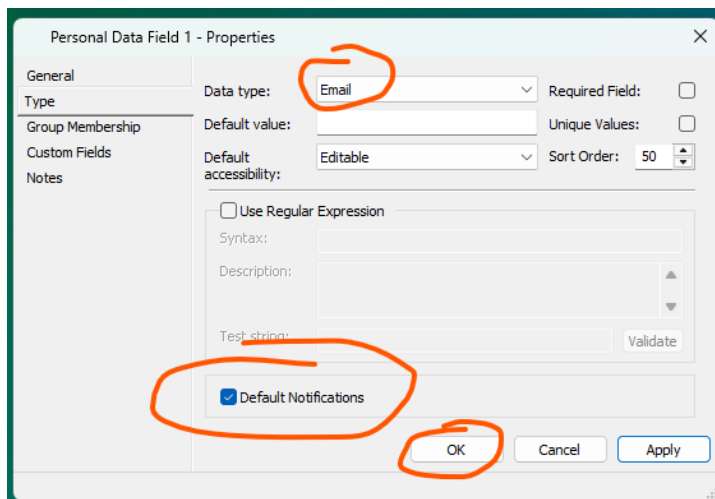
1. Create Personal Data Fields

The bridge application retrieves personal data field definitions during setup to enable you to map fields from OnLocation with Gallagher Command Centre. Any information you wish to sync will need to have a Personal Data Field preconfigured for the information to map to. The Name fields (first name, last name) are already available as default fields, but any other information such as email address, phone number, company, category etc, will need a Personal Data Field configured to hold this information for each cardholder record. Example – create email personal data field:

From Command Centre Configuration Client, click Configure → Personal Data Fields. Right click in the PDF window and select New → Personal Data Field. Add a Name (names will be pulled through to the bridge app for identification when creating matching rules). Ensure the Division is set to a division which the API will have access to (avoid using Root divisions to reduce access scope for the integration).

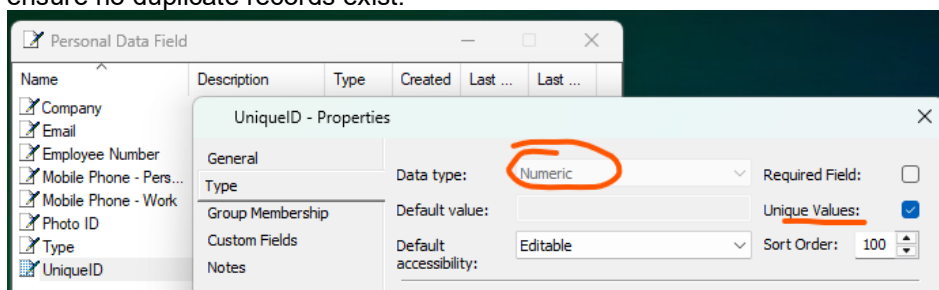


Set the type to email – this enables the PDF to be used as a notification address for cardholders. If you intend to use this field to notify cardholders, check the Default Notifications box too, then click OK:



Note: When setting restrictive data types, such as text-list, or setting Unique Values, attempting to push invalid data from the bridge app will fail, resulting in failed records within the bridge sync records which will need to be manually fixed.

Once you have created the Personal Data Fields you require, create a PDF to contain the Unique ID if you do not already have one. This should be a Numeric type PDF, and Unique Values must be checked to ensure no duplicate records exist:

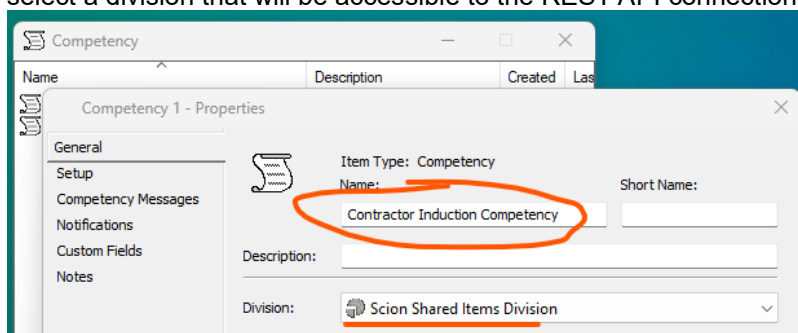


Optional: Create a PDF of text type which is used by the bridge interface to leave status messages against the cardholder record. The field will be updated with the latest activity by the bridge sync service to help trace events later if necessary.

2. Create Competencies for Inductions

The bridge application assumes the inclusion of induction data as the main driver toward integrating OnLocation with Command Centre. It is possible to skip this step if the only requirement is for synchronising of cardholder data from OnLocation. Example – create a Contractor Induction competency:

In Command Centre Configuration Client, open Configure → Competencies, right click in the Competencies window and select New → Competency. Enter a name to identify the competency, and select a division that will be accessible to the REST API connection.

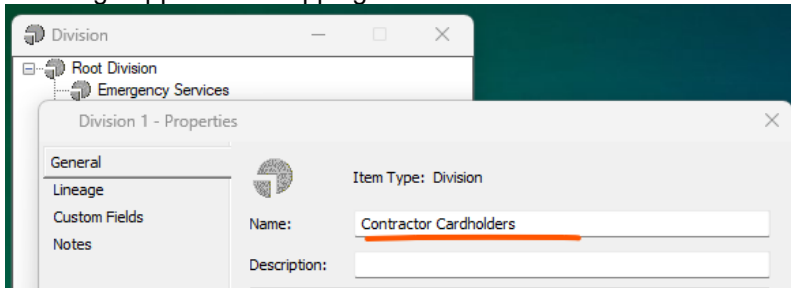


Configure the other tabs as required for the competency, note that the expiry date will be entered by the OnLocation induction expiry date, so no default expiry is required – but can be entered in case you need this for manual competency application. Click OK to save when finished.

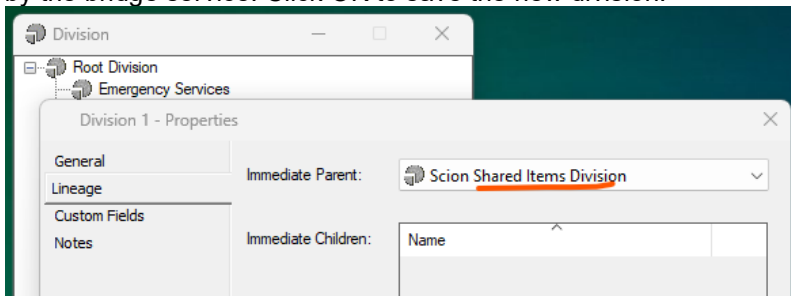
3. Create a Division for new cardholders

When the bridge application creates new cardholders in Command Centre from new records found in OnLocation, it requires a Division for the cardholders to be created in. As in previous steps, the division needs to be accessible to the API, so it is recommended not to use the site Root division. To create the Cardholder division:

In Command Centre Configuration Client select Configure → Divisions, right click in the Divisions window and select New → Division. Enter a name for the new Division – this will be the name it is identified by in the bridge application mapping section:



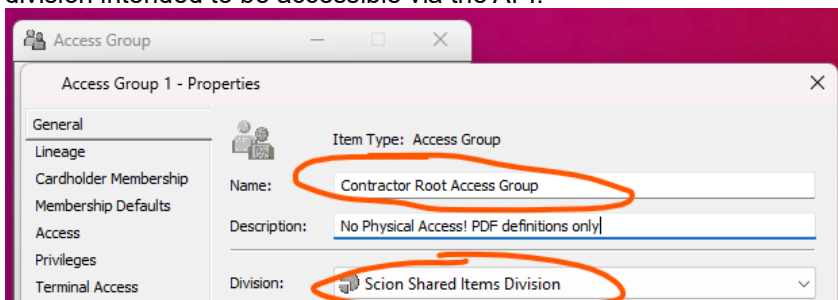
Select a parent division in the Lineage tab, it is suggested to nest the cardholder division underneath a division intended to group personal data fields, competencies and access groups which will be accessed by the bridge service. Click OK to save the new division:



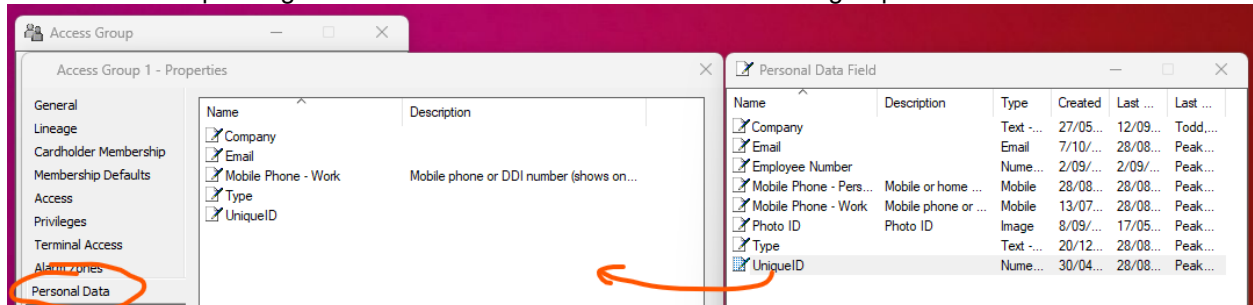
4. Create Access Groups for default access, and personal data field visibility

Access groups determine what card access the cardholders receive, but they also determine which personal data fields are available to the cardholders. The bridge application can be setup to apply a default access group or groups to newly created cardholders. This can be useful to apply access which is always relevant to all of the cardholders in the sync group. But it is also important that new cardholder records are linked to the relevant personal data fields. If no default access is required, then an access group can be created with no physical access, but containing the relevant personal data fields. Example – Contractor root access group for personal data definition:

In Command Centre Configuration Client, select Manage → Access Groups, then right click in the Access Groups window and select New → Access Group. Give it a descriptive name and set the Division to a division intended to be accessible via the API:



Set the lineage if required, noting that child groups inherit all access their parent groups have. Select the Personal Data tab and, with the Personal Data Fields window open, drag and drop the required fields into the Access Group dialog. Click OK to save and close the new access group:

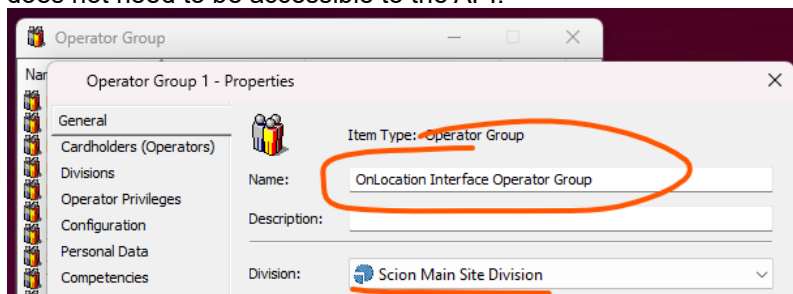


Note: a Unique ID personal data field is required, as it provides the matching field for ongoing sync between the two systems.

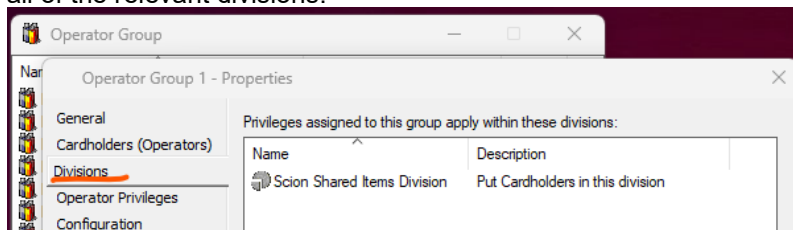
5. Create a REST API operator configuration and user to define permissions

The Gallagher Command Centre REST API access restrictions are defined by an operator – a cardholder with operator privileges. An Operator Group defines the access level and divisions accessible, and cardholder provides a traceable entity to provide the REST Client with access parameters, and log entry entity so that API events are traceable.

In Command Centre Configuration Client select Manage → Operator Groups, right click in the Operator Groups window and select New → Operator Group. Give the group a Name and select a division – this does not need to be accessible to the API:



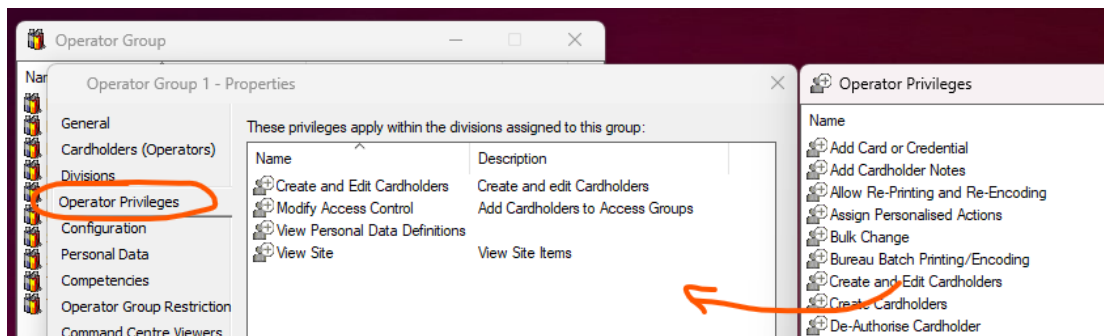
Select Divisions, and from the Divisions window drag and drop all relevant divisions (divisions for Cardholders, Personal Data Fields, Competencies and Access Groups – or a parent division containing all of the relevant divisions):



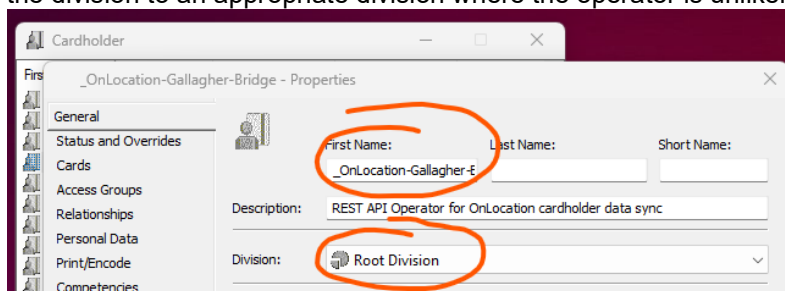
Select Operator Privileges, then open Manage → Operator Privileges. Drag and drop the following privileges into the Operator Group privileges dialog:

- Create and Edit Cardholders
- Modify Access Control
- View Personal Data Definitions
- View Site

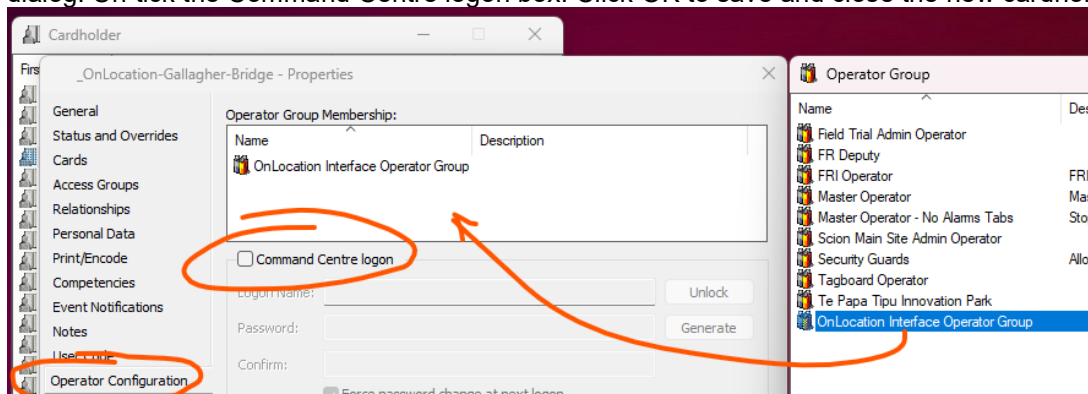
Click OK to save and close the new operator group



Create a new cardholder to act as the REST API operator, select Manage → Cardholders, right click in the cardholders window and select New → Cardholder. Add a meaningful name and description, and set the division to an appropriate division where the operator is unlikely to be accidentally removed or edited:

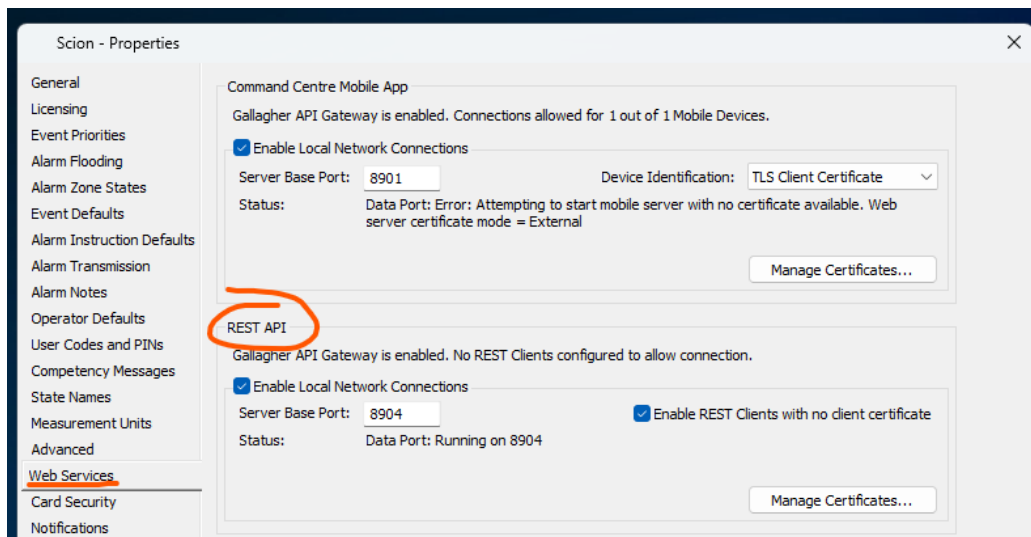


Go to the Operator Configuration tab, then drag and drop the new API operator group into the cardholder dialog. Un-tick the Command Centre logon box. Click OK to save and close the new cardholder:



6. Enable REST API and apply certificate

The REST API is disabled by default, so if your Command Centre installation has not previously been configured for REST API interfaces you will need to enable it and apply a trusted certificate. To enable the REST API system, in Command Centre Configuration Client click File → Server Properties and open the Web Services tab. Check Enable Local Network Connections, and if you are unable to provide a trusted certificate for the domain then check Enable REST Clients with no certificate:



If possible, create a certificate – add this by clicking Manage Certificates... set to Custom Certificate, and import your trusted certificate.

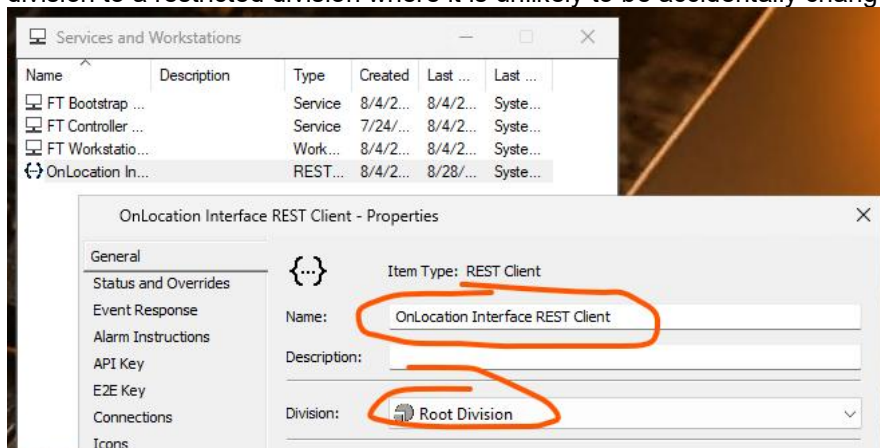
More info: the Gallagher Command Centre REST API requires a server certificate in **PKCS#12/PFX format (.pfx)**, including the associated private key.

The certificate must:

- Be issued by a Certificate Authority trusted by the systems that will connect to the Gallagher REST API.
- Contain the Gallagher server's fully qualified domain name (FQDN) in the **Subject Alternative Name (SAN)** field.
- Have **Server Authentication** enabled.
- Include the associated private key.
- Be supplied in PKCS#12/PFX format.
- Have an appropriate validity period in accordance with organisational certificate policy.

7. Create REST Client

The final step in the Gallagher Command Centre configuration is to create the REST Client, this assigns the operator and provides an API Key which will be required when setting up the bridge application. To create the REST Client, select Configure → Services & Workstations, right click in the Services and Workstations window and select New → REST Client. Give the client a meaningful name and set the division to a restricted division where it is unlikely to be accidentally changed or removed:



Select the API Key tab, then select the Rest Client Operator you created previously. Ensure that the Enable REST Client box is ticked, then hit Apply to save:

OnLocation Interface REST Client - Properties

General
Status and Overrides
Event Response
Alarm Instructions
API Key
E2E Key
Connections
Icons
Custom Fields
Notes

API Key:

Client Certificate Thumbprint:

REST Client Operator:
ADMIN, Contractor

Refresh Operator privileges

☒ Enable REST Client

REST privileges are cached and will not automatically update when an operator's privileges change. Be aware that refreshing operator privileges will interrupt any calls that are in progress.

Disabling a REST API Client can be a useful short term correction when an integration is causing a major performance problem with Command Centre.

OK Cancel Apply

Normal.

Configure OnLocation – Gallagher Bridge integration

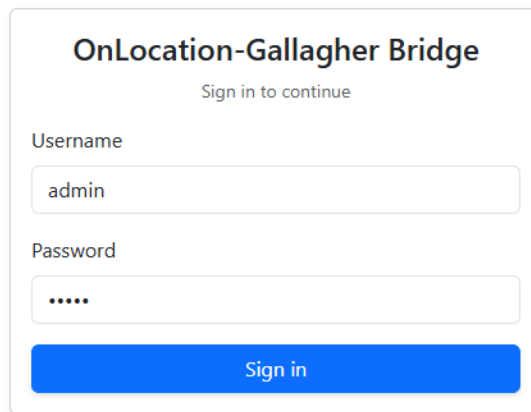
This is the final step in the process, it involves:

- Configuring the connections to both OnLocation and Gallagher APIs
- Mapping the data fields between systems and setting the unique ID for ongoing sync
- Completing the initial record match to match all existing records
- Starting automatic sync for ongoing system synchronisation.

Access the bridge application web UI through your browser, either on the local machine where you installed the bridge application, or from another PC with access to the bridge application url. On the local machine, the address to access initially is <https://localhost:5000> – remotely this would be <https://bridge-app-server-name:5000>.

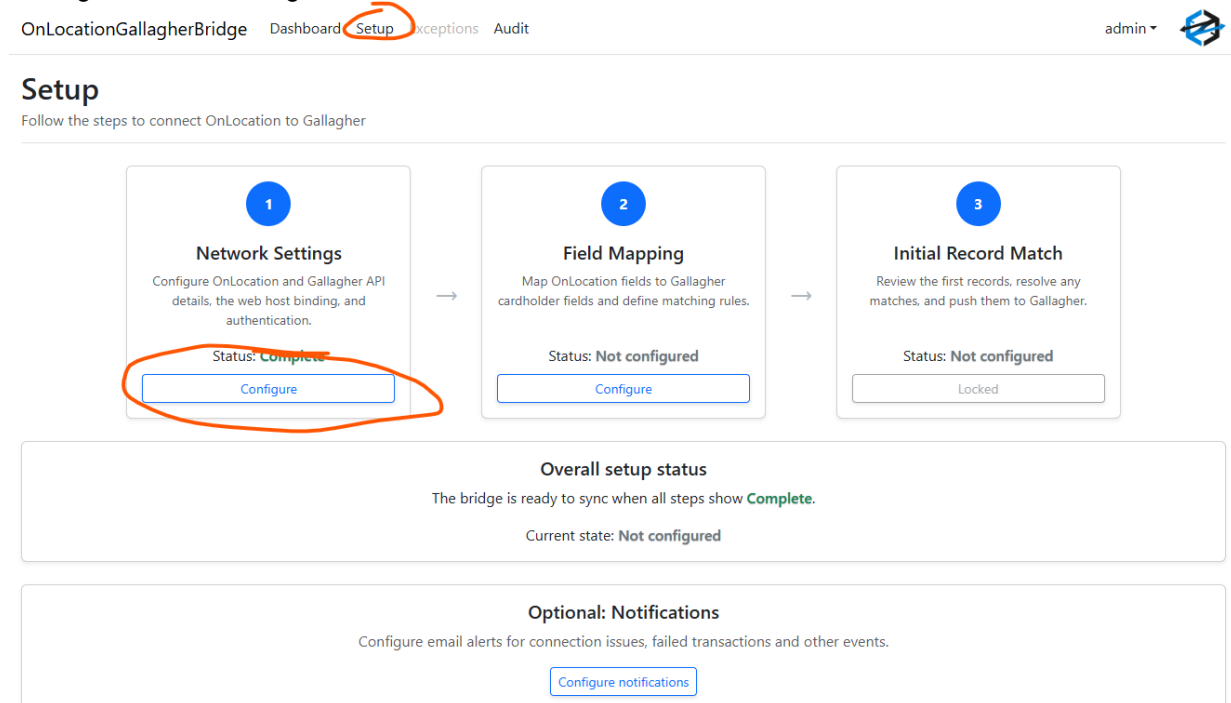
Note: the initial install uses a self-signed certificate, you will need to acknowledge your browser warnings to access the login page.

The sign in window appears, the initial username is **admin** and password is **admin**. You will be prompted to change the password for the first login, passwords must contain upper and lower case, numbers and symbols and must be at least 12 characters long – this is configurable once logged in as an administrator:



The login form is titled "OnLocation-Gallagher Bridge" with the subtitle "Sign in to continue". It contains two input fields: "Username" with the value "admin" and "Password" with masked characters ".....". A blue "Sign in" button is at the bottom.

The dashboard appears, showing OnLocation and Gallagher connections in a Fail state – as they are not yet configured. Click the Setup menu button at the page top to start setup, start with step 1 – Network Settings, click the Configure button:



The dashboard header shows "OnLocationGallagherBridge" and navigation links: "Dashboard", "Setup" (circled in orange), "Exceptions", and "Audit". The user "admin" is logged in.

Setup

Follow the steps to connect OnLocation to Gallagher

1

Network Settings

Configure OnLocation and Gallagher API details, the web host binding, and authentication.

Status: **Complete**

Configure

2

Field Mapping

Map OnLocation fields to Gallagher cardholder fields and define matching rules.

Status: **Not configured**

Configure

3

Initial Record Match

Review the first records, resolve any matches, and push them to Gallagher.

Status: **Not configured**

Locked

Overall setup status

The bridge is ready to sync when all steps show **Complete**.

Current state: **Not configured**

Optional: Notifications

Configure email alerts for connection issues, failed transactions and other events.

Configure notifications

Network Settings configuration

Enter your OnLocation and Gallagher API credentials and the Gallagher URL – you can test after entering the credentials, or just click Save and test connections which will test both and provide the test results. If one or both connections fail, you will need to correct this before moving on to the next stage of configuration. Common issues are certificate errors, if you do not have a valid certificate for the Gallagher API, check Disable TLS verification and test again. If the settings test OK, click the Next step button:

OnLocationGallagherBridge Dashboard Setup Exceptions Audit

admin



Network Settings

Configure how the bridge connects to OnLocation, Gallagher, and the local web interface.

Settings saved. Both connections verified.

Network Settings Complete This section is fully configured.

1 → 2

Save & test connections

Next step

Field Mapping

The bridge service retrieves the fields from both OnLocation and Gallagher APIs, this ensures that if permissions or API definitions change an up to date set of fields can always be retrieved from both systems:

Retrieving OnLocation profile records

1 OnLocation profile records

2 OnLocation inductions and holder records

3 Gallagher cardholder fields, divisions and access groups

This can take a little while. Keep this page open while the field data is retrieved.

If you have correctly configured the Gallagher Operator permissions then mapping should be straight forward. However if you find you are missing fields then check the operator group divisions, and which division each of the required entities is in. Each entity pulled through must be in a division that is available to the REST operator group, or a child division of the REST operator group divisions. Select a division for new cardholders, and any default access groups. Set the default action for records retrieved from OnLocation which do not exist in Command Centre – generally you will want to set this to Create new cardholder. Manual review will require operator intervention for any new records:

Defaults for new cardholders

Applied only when the bridge creates a cardholder. Existing cardholders keep the division and access groups they already have.

Division *

Contractors Division

Command Centre rejects a cardholder without a division, so this must be set before any cardholder can be created.

Access groups

Bike Shed Access
Canopy Tenant Access
Contractor General Access
DG Storage
E Block Fire Door
F Block Fire Door
Facilities Contractors
General Access Groups - NON CONTRACTORS ONLY

Optional. Hold Ctrl (or Cmd) to select more than one, or none to leave new cardholders without access.

Default action for unmatched records

Create new cardholder

What the bridge should do when a synced OnLocation record has no matching Gallagher cardholder.

Note: if you choose not to set any default access groups, no personal data fields will be visible to the interface, and this will cause sync to fail as the interface uses a unique ID PDF to match records during automatic sync. If you do omit this step, you will need to manually add an access group to new cardholders so that the personal data fields are accessible.

Next, set the Bridge sync message target field – you can use a purpose made PDF for this, the cardholder description field, or none.

Note: if no personal data fields appear in the list, check the Division that you placed these.

The next part of the page shows the available fields retrieved from both systems, including some sample data against each field for context. This is to aid in matching fields, as some field names from OnLocation are not obvious. Read through these to identify the relevant fields, if a field you expected to see is not there you can turn on all fields at the top of the page via the Show all OnLocation fields switch.

Once you have identified all the fields needed, start the mapping process below in the Field maps section. Some notes about transform functions (right hand column in the field mapping section):

- Choose Copy value where no change to the data or data type is required.
- For the name fields, OnLocation holds the first and last names in a single field, whereas .Command Centre separates these to first name and last name. Use the first name and last name transforms to split the OnLocation name field. First name grabs the first word in the OnLocation field, and last name copies everything after the first word.
- Use Gallagher competency expiry date to ensure correct date formatting between the OnLocation induction renew field and the Gallagher competency expiry field.
- Use rule based to set a Gallagher field name based on the contents of an OnLocation field. This could be a fixed name or value, or another OnLocation field value.

Example field map configuration:

Field maps

Map an OnLocation source field to a Gallagher target field. Choose **Rule based** to set the target only when conditions match.

OnLocation source field	Gallagher target field	Transform
id	UniqueID	Copy value
name	firstName	First name (first word)
name	lastName	Last name (remaining words)
email	Email	Copy value
mobile	Mobile Phone - Work	Copy value
Bioeconomy Science Institute – Scion Group Contractor Induction (ID 433) · learner_frc	Company	Copy value
id	Type	Rule based

Rule logic

All rules must match (AND)

Rules

1 IF id exists

Add rule

Remove

Output

Set to Contractor - Facilities

Value is a source field

Value is a number

Bioeconomy Science Institute – Scion Group Contractor Induction (ID 433) · renew

Contractor Induction Competency (expiry

Gallagher competency expiry da

Add field map

Maximum 30 field maps.

Next set the primary match field, this should be the OnLocation id field and your Command Centre unique ID personal data field. Set this to exact matching:

Primary match

The first rule used to identify the same person across OnLocation and Gallagher. Select more than one field on either side to compare a combined value, for example OnLocation **name** against Gallagher **firstName** and **lastName**.

OnLocation fields	Gallagher fields	Match type
Record group email id mobile name Bioeconomy Science Institute – Scion Group Contractor Induction (ID 433)	Mobile Phone - Personal Mobile Phone - Work OnLocation Interface Message Photo ID Type UniqueID	exact

Fallback match rules are the final step, and they aid with the initial matching stage next. You can select up to four match fields, and each field can match with multiple fields. Choose fuzzy as the match type to allow for minor differences such as case changes or minor spelling mistakes. Set the priority to greater than 0 to enable the rule:

Fallback match rules

When the primary match or existing ID mapping does not find a result, try these rules in priority order. Hold Ctrl (or Cmd) to select several fields on either side; multiple fields are joined with a space before comparing, so a single combined OnLocation field can be matched against separate Gallagher fields.

OnLocation fields	Gallagher fields	Match type	Priority
Profile record email id mobile name Bioeconomy Science Institute – Scion Group Contractor Induction (ID 433)	Cardholder authorised firstName lastName Personal Data Fields Company	fuzzy	1
Profile record email id mobile name Bioeconomy Science Institute – Scion Group Contractor Induction (ID 433)	Personal Data Fields Company Email Employee Number Mobile Phone - Personal Mobile Phone - Work OnLocation Interface Message	fuzzy	2
Profile record email id mobile name Bioeconomy Science Institute – Scion Group Contractor Induction (ID 433)	Personal Data Fields Company Email Employee Number Mobile Phone - Personal Mobile Phone - Work OnLocation Interface Message	fuzzy	3
Profile record email id mobile name Bioeconomy Science Institute – Scion Group Contractor Induction (ID 433)	Cardholder authorised firstName lastName Personal Data Fields Company	exact	0

Once all steps are completed, the click the Save mapping button at the top of the page, then the next step is enabled. Click next step once you are ready:

Mapping and match rules saved.

Field Mapping Complete This section is fully configured.

Record Group

Contractors

Refresh Remote Fields

Save mapping

Next step

Initial Record Match

Important: before completing this step you should back up the Command Centre configuration! This step can make very messy changes if there are problems with your configuration. **Assume you will not get this right first time and create a backup.**

OnLocation has separate API calls for Staff and Contractors, so these are available as separate profiles in the bridge application. Select the record group you wish to map, in this case well select Contractors.

Next click Load inductions for this record group – only relevant inductions will be loaded, this avoids looking for irrelevant inductions for the selected group.

Tick the induction/s you want to retrieve, then set the induction period to search. Rather than retrieving all historical records you may wish to only pull through the most recent – if your inductions typically only last one or two years then this is probably an appropriate period to set. If an older inductee who is out of date later completed the induction refresher, the sync process will add them to Command Centre once their induction has been completed.

Finally select the default action for records which have no matching data in Command Centre – you will most likely want to select Create new. Note that you will have a chance to change the action at an individual record level before starting the initial match push through to Command Centre.

Once these fields are completed, click the Build initial match review button:

Initial Record Match

Initial Record Match Incomplete Some required configuration is still missing.

Record Group

Contractors

Load inductions for this record group

Inductions to pull through
Only ticked inductions are scanned, for the initial match and for ongoing sync. Each unticked induction saves a full holder scan.
☒ Bioeconomy Science Institute – Scion Group Contractor Induction (id 433)

Include induction records completed within the last

730

days

Default resolution for records without an automatic match

Create new

Build initial match review

Go to dashboard

This may take some time as the interface retrieves and compiles the records from both systems, attempting to automatically match records.

A summary of the records retrieved and matched appears, with the entire list of records below. If the match system was unable to match records, a count of these shows in the Review and confirm summary window. The unresolved records appear at the top on the list, work through these to manually resolve them. You can expand the fields for each person for more information if you need to correlate based on other data (such as Company for example).

Once all record matches are resolved with an action, the **Confirm initial match and enable sync** button is enabled:

Review and confirm

Resolve every OnLocation record before confirming. Records that still need attention are shown at the top.

675

OnLocation records in window

1581

Gallagher cardholders

0

Skipped (no usable name)

0

Need attention

269

Match

403

Create

3

Exclude

All records have a solution. You can now confirm and enable sync.

Confirm initial match and enable sync

Click **Confirm initial match and enable sync**, a final warning to backup appears – ensure you complete the backup if you have not already! Click the acknowledgment check box then Confirm and enable sync. New records begin pushing through to Command Centre, you can monitor this via Command Centre event view. Once the initial matching is complete, the first sync is scheduled and will begin about 1 minute after the initial push. Head back to the Dashboard and check on any failed records. Any failed records can be fixed manually in the Exceptions page. If the exceptions continue to fail, more information about why the update is failing will show in the Audit log or recent activity list on the Dashboard. You may need to adjust settings or existing records in OnLocation or Gallagher Command Centre to fix some issues.